

A 0D pulsatile model of aortic-arch banding

Reproducing carotid pulsatility redistribution (Eberth et al., 2009)

svZeroDSolver

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The problem

Eberth et al. (2009): a stiff band on the mouse aortic arch, placed *between* the two carotid takeoffs, creates paired carotids with **similar mean pressure & flow but very different pulsatility**.

- **RCCA-B** (right) — *upstream* of the band: sees the full pulse + reflections $\Rightarrow$ pulsatility index rises ($PI \approx 3.11$).
- **LCCA-B** (left) — *downstream*: the resistive–inertial band damps the pulse ($PI \approx 1.65$); baseline CCA ≈ 1.16 .

Goal: a reduced-order (0D) lumped model that reproduces this redistribution with the band as the only structural change — and that maps to the wall growth & remodeling Eberth measured.

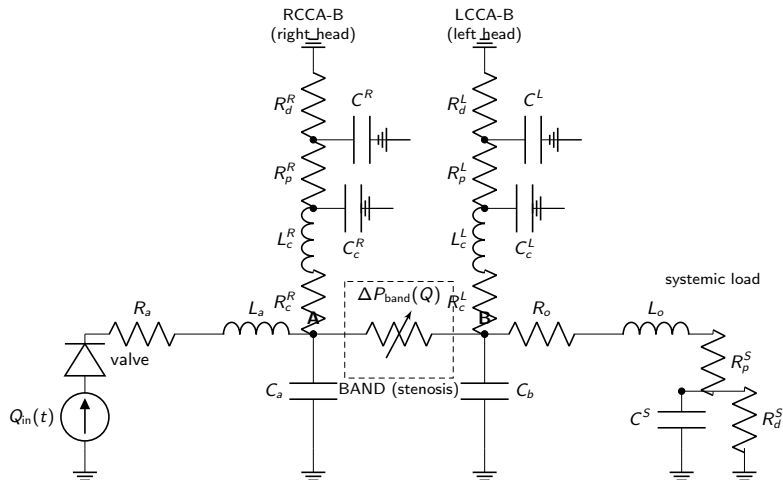
The 0D model

Heart $\rightarrow$ aortic valve $\rightarrow$ ascending aorta (R_a, L_a, C_a) $\rightarrow$ **node A** $\rightarrow$ band $\Delta P_{\text{band}}(Q) \rightarrow$ **node B** $\rightarrow$ descending aorta (R_o, L_o, C_b); each node feeds a carotid vessel (R_c, L_c, C_c) into a terminal RCR bed (R_p, C, R_d).

- Vessels: RCL BloodVessel blocks
- Band: nonlinear *Young-Tsai* stenosis
$$\Delta P_{\text{band}} = (R_{\text{band}} + S|Q|)Q + L_{\text{band}} \dot{Q}$$
- Terminal beds: 3-element RCR windkessels
- Valve: ValveTanh diode
- Parameters from mouse literature (geometry, blood, wall stiffness E)
- Every parameter tagged
FIXED / DERIVED / CALIBRATE
- Only the band `stenosis_coefficient` is calibrated (to the Table-1 PI ratio)

Three states differ *only* in carotid geometry: pre-banding (baseline), acute (baseline walls), chronic (Table-1 remodeled walls).

0D circuit (electric analogy)

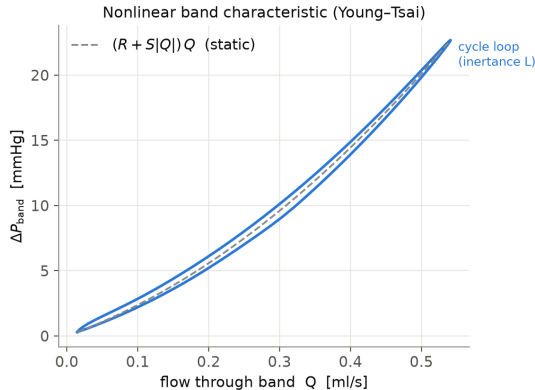


symbol	component (this repo)
$Q_{in}(t)$	INFLOW
valve	aortic_valve
R_a, L_a, C_a	ascending_aorta
$\Delta P_{band}(Q)$	aortic_band
R_o, L_o, C_b	descending_aorta
node A / B	node_A / node_B
R_c^R, L_c^R, C_c^R	rcca_b
R_p^R, C_c^R, R_d^R	RCR_RIGHT
R_c^L, L_c^L, C_c^L	lcca_b
R_p^L, C_c^L, R_d^L	RCR_LEFT
R_p^S, C^S, R_d^S	RCR_SYS

Not drawn (negligible): aortic_root (R, L before the valve). Note: $C_c^{R,L}$ (the carotid's own wall compliance, $\sim 1e-6$ ml/mmHg) is $\sim 30-40\times$ smaller than the bed $C^{R,L}$ it feeds into — it barely affects PI, which is why wall remodeling alone doesn't move it (see next slide).

Pressure $\leftrightarrow$ voltage, flow $\leftrightarrow$ current. Node A (RCCA-B) is upstream of the band and stays pulsatile; node B (LCCA-B) is downstream and damped. Each terminal bed is an R_p-C-R_d Windkessel.

The band is a nonlinear (Young–Tsai) element



ΔP – Q across the band: the quadratic Bernoulli/turbulent term ($S|Q|Q$) plus an inertance loop ($L \dot{Q}$). This nonlinearity + inertia is what damps the downstream pulse while leaving the upstream node pulsatile.

Parameter roles: fixed, assumed, and the knob we tune

Fixed (measured / literature)

- blood $\rho = 1.06$, $\mu = 3.5$ cP
- wall E : aorta 1.0, carotid 1.5 MPa
- HR 6.09 Hz, CO 0.20 ml/s, MAP 90
- carotid ID/wall (Table 1), aorta dia.
- measured central aortic compliance

⇒ all vessel R , L , C are *derived* from these via Eq. (1)–(3).

Assumed (plausible, not fitted)

- segment lengths (carotid, band) — estimates
- ejection waveform shape (systolic fraction)
- RCR split $R_p:R_d = 10:90$; time const. τ
- distal pressure $P_d = 0$
- valve R_{max}/R_{min} /steepness

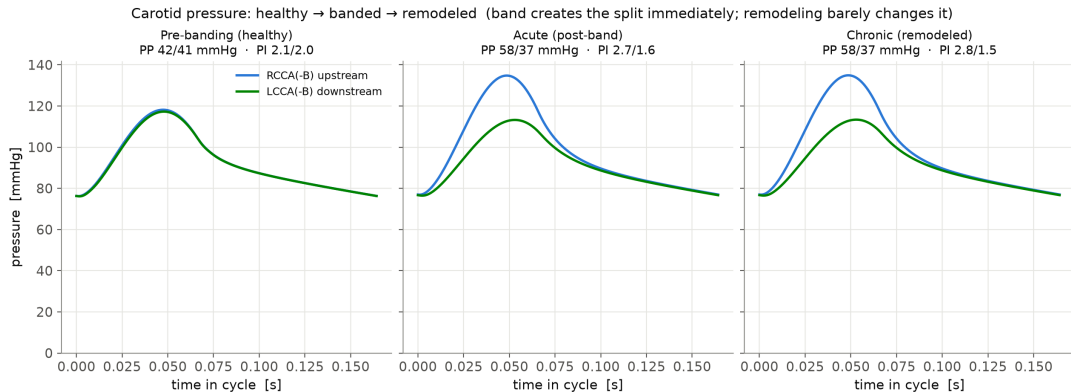
chosen by convention/estimate — could be refined, but not tuned to the data.

Knob (calibrated to data)

- band stenosis_coefficient S
- the **one** parameter we turn
- geometry gives a prior; tuned to the Table-1 PI ratio (3.11/1.65)

Everything else is fixed or derived — this is the only free knob fitted to the data.

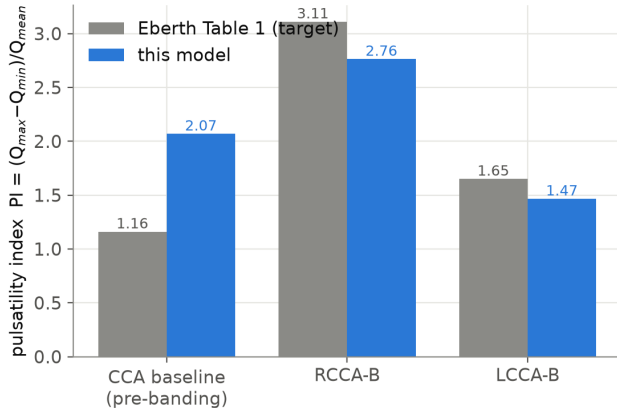
Result: banding creates the split immediately



Carotid pressure is **symmetric before banding**, splits **immediately** on banding (upstream $\uparrow$, downstream damped), and **barely changes** with wall remodeling $\Rightarrow$ the split is set by band *position*; remodeling is the response, not the cause.

Result: pulsatility vs. the paper

Pulsatility by location — calibrated model vs paper Table 1
(stenosis calibrated to the PI ratio; both PIs within ~7%)



	Paper	Model
RCCA-B PI	3.11	≈ 2.8
LCCA-B PI	1.65	≈ 1.5
PI ratio	1.89	1.89
PP RCCA-B	56	58
MAP (R/L)	similar	98/92

The **banded** carotids, the PI ratio, MAP similarity, and pulse pressure are reproduced; mean carotid flows land in the measured range.

Limitation: baseline vs. banded cannot both be matched in 0D

- The model **over-predicts the baseline (pre-banding) PI** (≈ 2.1 vs. the paper's 1.16), though baseline *pulse pressure* matches (≈ 41 vs. 42).
 - A parameter sweep shows every knob that lowers the baseline PI to 1.16 **also collapses the banded PI** — the passive bed is a linear pulsatility filter.
 - The paper's baseline→banded amplification is $\sim 2.7\times$ ($1.16\rightarrow 3.11$); a lumped 0D band delivers only $\sim 1.5\times$.
 - The missing $\sim 2\times$ is **wave reflection** off the band — a spatial/timing effect a 0D element cannot represent.
- ⇒ Reproducing the *full* amplitude needs a **1D** (wave-propagation) model — exactly the 0D→1D boundary.

Targets: pinned, fixed, or an actual free knob?

Control group (CCA) — 1 free knob

target	how it's matched	status
$\bar{Q}=0.016$	bed R_p+R_d	pinned
MAP=92	sys $R_p^S+R_d^S$	pinned
PP=42	C_a, C_b (aortic)	emergent
PI=1.16	bed $C^{R,L}$	KNOB

Banded group (RCCA-B / LCCA-B) — 3 free knobs

target	how it's matched	status
$\bar{Q}=0.022/0.012$	bed $R_p^R+R_d^R / R_p^L+R_d^L$	pinned
MAP=86/77	sys R (pinned) + band S	KNOB
PP=56/27	C_a, C_b (aortic)	emergent
PI=3.11/1.65	bed C^R / C^L	KNOB

Only 4 parameters are actually free/calibrated (control bed $C^{R,L}$; banded bed C^R, C^L , band S). Bed resistances (R_p+R_d , all beds) are *pinned* — solved directly from $(\bar{Q}, \text{MAP})$ via Ohm's law, not searched. PP is matched by the aortic C_a, C_b , which is *fixed* literature identical in both groups — an *emergent* match, not a tuned one. Also fixed: HR (7.17/6.09), CO, carotid geometry, blood μ, ρ , wall moduli E , ejection shape.

How each parameter is actually derived

1) Bed resistance R_p+R_d — closed form, not searched. Time-average the RCR equations over one period ($\overline{dP_c/dt} = 0$ in periodic steady state):

$$P_{in} - P_c - R_p Q_{in} = 0, \quad R_d Q_{in} - P_c + P_{ref} - R_d C \dot{P}_c = 0 \xrightarrow{\text{avg}} R_p + R_d = \frac{\text{MAP} - P_{ref}}{\bar{Q}} = \frac{\text{MAP}}{\bar{Q}} \quad (P_{ref}=0)$$

using each vessel's own measured $\bar{Q}$ and target local MAP; split $R_p:R_d = 10:90$ (assumed convention). Same formula for RCR_RIGHT/LEFT/SYS in both groups.

2) Bed compliance C — bisected against PI (monotone in C). Low C : capacitor inert, PI $\rightarrow$ PP/MAP (resistive limit); high C : capacitor adds flow swing, PI grows. 28-step bisection on $C \in [10^{-7}, 2 \times 10^{-4}]$ until simulated PI = target:

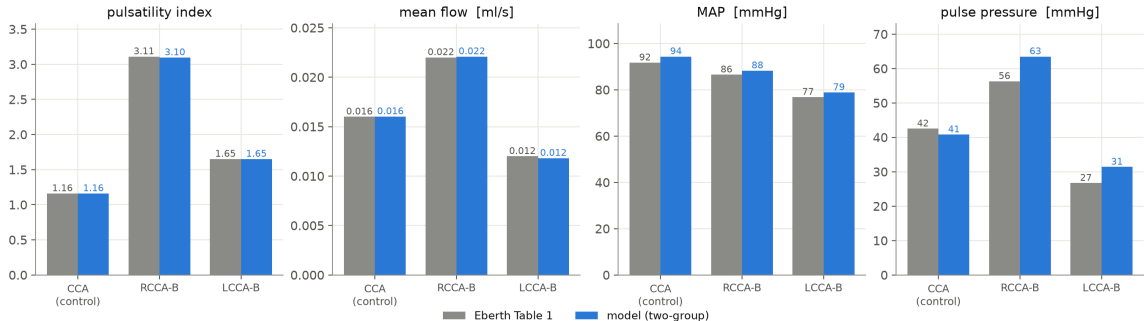
	target PI	solved C (ml/mmHg)	script
control $C^{R,L}$	1.16	6.44×10^{-6}	make_control_group.py
banded C^R (RCCA-B)	3.11	2.56×10^{-5}	make_banded_group.py
banded C^L (LCCA-B)	1.65	2.33×10^{-5}	make_banded_group.py

3) Band S — bisected against the measured $A \rightarrow B$ pressure drop. Raising S increases the drop monotonically; bisect $S \in [20, 600]$ until $\text{MAP}_A - \text{MAP}_B = 9.7$ (Table 1) $\Rightarrow S = 95.1$.

Revised: two-group calibration (control vs. banded)

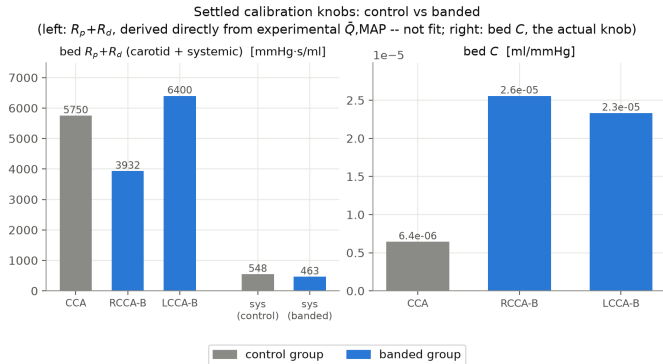
Control (CCA) and banded (RCCA-B/LCCA-B) are **two separate animal groups** — they need not share $R/C/L$. Calibrating each on its own terms — HR and measured flows/pressures pin the beds, the band anchored to the measured $A \rightarrow B$ pressure drop, per-carotid bed compliance sets the PI — matches *all three* carotid states with a stock RCR:

Two-group calibration vs Eberth Table 1 (control CCA + banded RCCA-B/LCCA-B)



Only **4 parameters are free** (bed $C^{R,L}$, C^R , C^L , band S — previous slide); bed R pinned, aortic C fixed. Cost: bed compliance differs between groups (fitted) — descriptive, not predictive.

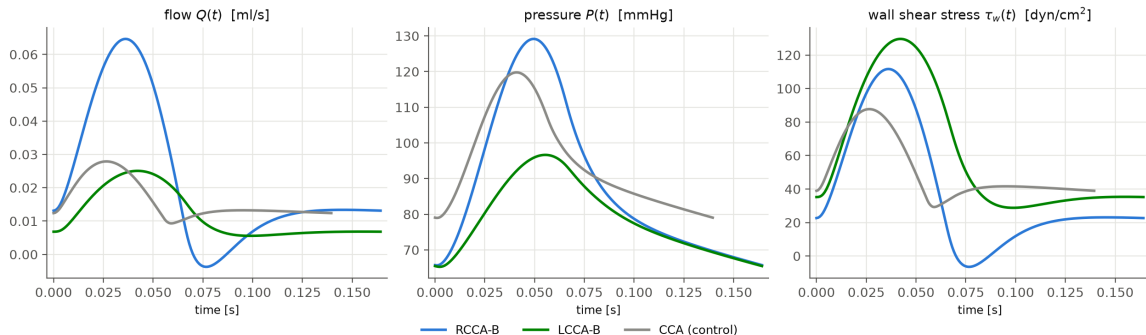
Settled calibration knobs: control vs. banded



$R_p + R_d$ (left, both carotid and systemic beds) is **derived directly from experimental data** ($\bar{Q}$, MAP via Ohm's law) — not fit. Bed C (right) is the one actual calibrated knob.

Looking ahead: inputs for growth & remodeling (G&R)

RCCA-B / LCCA-B (banded) vs. CCA (control) time series, each over its own converged cycle



Flow $Q(t)$ and pressure $P(t)$ come directly from the solver; wall shear stress $\tau_w(t)$ is **not** a solver output — computed via Poiseuille $\tau_w = 4\mu Q(t)/(\pi r^3)$ using each vessel's fixed lumen radius. These per-cycle time series (not just the mean/PI summaries used for calibration) are the raw inputs a future G&R model would consume.

Summary

- A literature-parameterized 0D model reproduces the Eberth **pulsatility redistribution**: upstream carotid pulsatile, downstream damped, at matched mean pressure/flow.
- Banding creates the split **immediately**; wall remodeling barely changes it — consistent with pulsatility being the *stimulus* for remodeling.
- With a *single shared* parameter set, 0D matches the redistribution and pressures but not both the baseline and banded PIs (a bounded amplification limit; a 1D wave-reflection effect is small for the mouse).
- Treating control and banded as *separate groups* — their own beds, band anchored to the measured pressure drop — matches **all three carotid states**, at the cost of group-specific (fitted) bed compliance.